Why Agentic Testing Framework

Competitive Differentiation

| Capability | Tr# Agentic Testing Framework

AI-Powered Quality Assurance

Transform Your QA Operations with Intelligent Automation

Slide 1: The Problem

Traditional Test Automation is Broken

The Maintenance Trap

- Script Fatigue: Test automation engineers spend 70%+ of time fixing scripts, not testing
- Rework Cycles: Every UI change triggers cascading script failures
- Talent Drain: Skilled engineers become script maintenance workers
- Coverage Gaps: Teams revert to manual testing due to automation overhead

The Hidden Costs

- Delayed releases due to regression bottlenecks
- Bug leakage to production despite automation investment
- Rising headcount without proportional quality improvement
- Technical debt accumulating in test suites

> *"We had a huge team of test automation engineers. All they were doing is not testing the application, but fixing the scripts."*

Slide 2: Our Solution

Agentic Testing Framework

Zero-Script, AI-Native Test Automation

From Natural Language to Executed Tests - No Code Required

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│                                                                 │
│   📝 Natural Language    →   🤖 AI Execution    →   📊 Results│
│      Test Cases                  Engine              Dashboard│
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Key Innovation

- Write tests in plain English - No Selenium scripts, no XPaths, no locators
- AI understands your application - Adapts to UI changes automatically
- Execute at scale - Kubernetes-powered parallel execution
- Intelligent reporting - Role-based insights for every stakeholder

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Slide 3: How It Works

The Intelligent Testing Pipeline

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│                      AGENTIC TESTING WORKFLOW
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│   │    AGENT 1     │     │    AGENT 2     │     │    AGENT 3     │
│
│   │  Test Case     │───▶│   Context      │───▶│ Execution      │
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│   │  Generation    │     │    Builder     │     │     Engine     │
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│
│   │ Analyzes PRDs│     │ Builds Smart   │     │  Runs Tests    │
│
│   │ & User         │     │ Prompts from   │     │  on Browser    │
│
│   │ Stories        │     │ Knowledge      │     │  via AI        │
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│   │                │     │ Base           │     │  Vision        │
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The Magic: No Script Maintenance

- AI "sees" the application like a human tester
- Automatically adapts to UI changes
- Self-healing test execution
- Continuous learning from each run

Slide 4: Architecture Overview

Enterprise-Ready Architecture

[PLACEHOLDER: High-Level Architecture Diagram]

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│  │                      INTEGRATION LAYER
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│
│  │  │   JIRA   │  │Confluence│  │ GitHub/ │  │ServiceNow│
│   │
│  │  │          │  │          │  │BitBucket│  │          │            │
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│  │                    INTELLIGENCE LAYER
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│
│  │  │  Knowledge  │  │   Context   │  │  Snapshot   │          │
│
│  │  │    Base     │  │    Layer    │  │  Manager    │          │
│
│  │  │ (RAG/Vector)│  │ (Prompting) │  │ (Page State)│          │
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│  │                    EXECUTION LAYER
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│  │  ┌─────────────┐  ┌─────────────┐  ┌─────────────┐          │
│
│  │  │Orchestrator │  │   Browser   │  │ Kubernetes  │          │
│
│  │  │   (LLM +    │  │ Automation  │  │ Scaling     │          │
│
```

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│   │   │     MCP)         │   │    Layer        │   │  Engine        │           │
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```
Screenshot Placeholder: Insert detailed architecture diagram
```

Deployment Flexibility

- ✅ AWS / Azure / GCP
- ✅ On-Premises
- ✅ FedRAMP-ready components
- ✅ Air-gapped environments supported

Slide 5: Dashboard - Executive View

C-Suite Intelligence Dashboard

```
[PLACEHOLDER: Executive Dashboard Screenshot]
```

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┌──────────────────────────────────────────────────────────────────────────
──────┐
│  Agentic Testing Dashboard                                 👤 Executive View
│
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```

Time to Market
↓ 40%

Automation ROI
$2.3M saved

Production Defect Rate
↓ 67%

Risk Reduction
89%

Quality Trend (12 months)
📈 [Chart]

Cost Savings Analysis
📊 [Chart]

🤖 AI Assistant: "Quality improved 23% this quarter. Key driver: 94% automation coverage on critical paths."

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Executive Metrics

- Time to Market - Release cycle acceleration
- Automation ROI - Cost savings from reduced manual effort
- Production Defect Rate - Quality improvement tracking
- Risk Reduction Score - Proactive risk identification

Slide 6: Dashboard - Manager View

QA Manager Operations Dashboard

```
[PLACEHOLDER: Manager Dashboard Screenshot - Reference
dtq.digitalworkplace.ai]
```

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│  Agentic Testing Dashboard                         👤 QA Manager View
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│  ┌──────────────┐ ┌──────────────┐ ┌──────────────┐
┌──────────────┐ │
│  │ Total        │ │ Automation   │ │    Risk      │ │    Open
│  │
```

```
│   │ Features       │ │    Rate        │ │ Distribution │ │  Defects
│   │
│   │    46          │ │    43%         │ │  3H/25M/18L  │ │     47
│   │
│   │ 87.8% avg      │ │ 20 automated │ │              │ │
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│
│   │  ⚠️ HIGH RISK FEATURES REQUIRING ATTENTION
│     │
│   │   ┌─────────────────┐ ┌─────────────────┐ ┌─────────────────┐      │
│
│   │   │xAPI/LRS Integ.  │ │Learning Analyt.│ │Mobile Experience│      │
│
│   │   │Coverage: 92%    │ │Coverage: 89%    │ │Coverage: 79%    │      │
│
│   │   │Risk: 45         │ │Risk: 42         │ │Risk: 42         │      │
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│
│   │  Test Pass Rate Trend      │  │   Feature Coverage Matrix       │
│
```

```
│   │       📈 94.2%                 │  │          📊 [Heatmap]              │
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Manager Metrics

Metric	Value	Trend
Test Pass Rate	94.2%	+2.3%
Automated:Manual Ratio	8.5:1	+0.5
First-Run Pass Rate	87.5%	+4.2%
Regression Duration	42 min	-8 min
Environment Uptime	99.2%	Stable

Slide 7: Dashboard - Tech Lead View

Tech Lead Execution Console

```
[PLACEHOLDER: Tech Lead Dashboard Screenshot]
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│  Agentic Testing Dashboard                          👤 Tech Lead View
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│  │  🚀 RUN REGRESSION TESTS
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│
│  │  │  Select Features:                                        │
│    │
│  │  │  ☑️ User Authentication    ☑️ Course Management           │
│    │
│  │  │  ☑️ Payment Processing     ☐ Reporting Module             │
│    │
│  │  │  ☑️ Mobile Experience      ☐ Admin Functions              │
│    │
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│
│  │
│    │
│  │  Environment: [Production ▼]    Browser: [Chrome ▼]             │
│
│  │
│    │
│  │  [▶️ EXECUTE TESTS]  [📋 VIEW QUEUE]  [⏸️ SCHEDULED RUNS]        │
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```
│
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│
│  │ 📊 CURRENT EXECUTION STATUS
│   │
│  │ ██████████████████░░░░ 74/100 Tests Complete       │
│
│  │ ⏱️ 3:42 elapsed | 🖥️ 20 parallel instances | ✅ 71 passed     │
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Tech Lead Capabilities

- On-Demand Regression - Trigger full or selective test runs
- Parallel Execution - Scale to 20+ concurrent instances
- Real-Time Monitoring - Live execution status and logs
- Immediate Results - Results pushed to dashboard automatically

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Slide 8: AI Assistant

Intelligent Quality Insights

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[PLACEHOLDER: AI Chatbot Interface Screenshot]

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│  🤖 AI Assistant
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│
│  │ 💬 Chat History
│    │
│  │ ───────────────────────────────────────────────────────────────
│    │
│  │
│    │
│  │ 👤 "What are the high-risk features this sprint?"
│    │
│  │
│    │
│  │ 🤖 "I've identified 3 high-risk features:
│    │
│  │     1. Mobile Experience (79% coverage, 3 open defects)
│    │
│  │     2. xAPI Integration (Risk score: 45)
│    │
│  │     3. Learning Analytics (2 regressions in last run)
│    │
│  │
│    │
│  │     Recommendation: Prioritize Mobile Experience testing
│    │
│  │     before Thursday's release."
│    │
│  │
│    │
│  │ 👤 "Show me the defect trend for Payment module"
│    │
```

```
│   │
│    │
│   │ 🤖 [📊 Displaying trend chart...]
│    │
│   │     "Payment module defects decreased 34% over 4 sprints.
│    │
│   │     Current open: 2 medium severity."
│    │
│   │ ──────────────────────────────────────────────────────────
│    │
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│    │
│   │ [Type your question...]                              [Send 📤]
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Ask Anything

- "What features have the lowest test coverage?"
- "Why did last night's regression fail?"
- "Compare quality metrics between releases"
- "What should we focus on before go-live?"

Slide 9: Key Features

Framework Capabilities

🧠 Intelligent Test Generation

- Auto-generates test cases from PRDs & user stories
- Learns from existing test suites
- Eliminates duplicate scenarios
- Human-in-the-loop approval workflow

🎯 Zero-Code Execution

- Natural language test cases
- No XPaths, locators, or scripts
- Self-healing against UI changes
- Works with any web application

⚡ Scalable Execution

- Kubernetes-powered parallel runs
- Auto-scaling based on demand
- 20+ concurrent browser instances
- Full regression in minutes, not hours

📊 Role-Based Analytics

- Executive: ROI & strategic metrics
- Manager: Operations & team performance
- Tech Lead: Execution & technical depth
- AI-powered insights for all

🔗 Enterprise Integration

- JIRA / Confluence
- GitHub / BitBucket
- ServiceNow
- CI/CD Pipelines (Jenkins, etc.)

Slide 10: Results & Impact

Proven Outcomes

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│                              CUSTOMER RESULTS
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│   │      100%        │  │      ZERO        │  │      94%+        │
│
│   │   REGRESSION     │  │   XPath/Script   │  │   Test Pass      │
│
│   │   Handed to      │  │   Knowledge      │  │     Rate         │
│
│   │   AI Agent       │  │   Required       │  │   Achieved       │
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│   │                  │  │                  │  │                  │
│
│   │      70%         │  │     ZERO         │  │    QA TEAM       │
│
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```
│  │   REDUCTION      │  │   UI Change      │  │   REFOCUSED      │
│
│  │   in script      │  │   Rework         │  │   on Sprint      │
│
│  │   maintenance    │  │   Required       │  │   Testing        │
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Before & After

Metric	Before dTQ	After dTQ
Regression Ownership	QA Team (manual)	AI Agent (autonomous)
Script Maintenance	70% of QA time	<10% of QA time
Test Automation Engineers	Fixing scripts	Actual feature testing
UI Change Impact	Days of rework	Zero rework
Bug Leakage	Frequent	Rare
QA Focus	Regression + Sprint	Sprint testing only

"Regression is completely handed over to this agent. QA engineers now focus on sprint testing, coordinating with tech leads, and retesting bug fixes."

Slide 11: Deployment Options

Flexible Deployment for Any Environment

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│  │  OPTION 1: CLOUD (AWS / Azure / GCP)
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│
│  │  │   dTQ      │  │   LLM      │  │    K8s     │  │ Vector     │              │
│
│  │  │Dashboard│  │ Service │  │ Cluster │  │    DB      │              │
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│
│  │  ✅ Fastest deployment  ✅ Auto-scaling  ✅ Pay-as-you-go      │
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│  │  OPTION 2: GOV CLOUD (AWS GovCloud / Azure Government)
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│
│  │  │  dTQ     │  │ Bedrock/│  │   K8s    │  │ Vector   │           │
│
│  │  │Dashboard│  │ Azure AI│  │ Cluster │  │   DB     │           │
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│
│  │  ✅ FedRAMP compliant  ✅ Approved LLMs  ✅ Data residency    │
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│  │  OPTION 3: ON-PREMISES / AIR-GAPPED
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│  │  │  dTQ     │  │Open-Src │  │   K8s    │  │ Vector   │           │
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│  │  │Dashboard│  │  LLM     │  │ On-Prem │  │   DB     │           │
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│
│  │  ✅ Complete data control  ✅ No external calls  ✅ Air-gap    │
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Security & Compliance

- Uses YOUR approved LLMs (no external API calls required)
- Data never leaves your environment
- Supports classified/sensitive workloads
- No FedRAMP/ATO required for framework itself

Slide 12: Implementation Approach

Getting Started with dTQ

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│                        IMPLEMENTATION PHASES
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│
│  PHASE 1                PHASE 2                PHASE 3
PHASE 4│
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```
│  │   AI    │          │Context │          │ Deploy │          │
Scale │││
│  │Readiness│   ───▶  │  Layer │   ───▶  │   &    │   ───▶  │   &
│││
│  │ Assess  │          │  Setup │          │Onboard │
│Optimize│││
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└─────────┘│
│
│
│  Week 1-2             Week 3-4            Week 5-6
Ongoing   │
│
│
│  • Data quality       • Knowledge base    • Deploy to env     • Add
│
│    assessment           configuration     • Train users
features │
│  • Document           • Integration       • Initial test      • Tune
│
│    standardization      setup               runs
prompts  │
│  • Readiness          • Prompt            • Go-live           •
Expand  │
│    report               engineering
coverage │
│
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AI Readiness Assessment

"If you're not ready to actually use an agent, you better not even buy it."

We evaluate:

- Documentation quality & consistency
- Existing test asset structure
- Integration touchpoints
- Team readiness

Slide 13: Why dTQ

Competitive Differentiation

Capability	Traditional Automation	Low-Code Tools	dTQ
Script Maintenance	High	Medium	None
UI Change Adaptation	Manual rework	Partial	Automatic
Learning Curve	Months	Weeks	Days
Execution Speed	Sequential	Limited parallel	Massively parallel
AI-Powered Insights	None	Basic	Deep intelligence
Deployment Flexibility	Tool-dependent	SaaS only	Any environment
Vendor Lock-in	High	High	Low (Framework)

Our Approach: Framework, Not Product

- Not a SaaS subscription - deploys in YOUR environment
- No vendor lock-in - uses open standards
- Fully customizable - adapts to your needs
- IP protection - your data stays yours

Slide 14: Let's Talk

Next Steps

See It In Action

Request a live demonstration with YOUR application

Proof of Value

- 4-6 week pilot program
- Measurable outcomes
- Low-risk evaluation

Contact

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│   🌐   [Your Website]
│
│   📧   [Contact Email]
│
│   📞   [Phone Number]
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│
│
│   Available on:
│
│   • Trade Winds Marketplace (Coming Soon)
│
│   • Direct Engagement
│
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Appendix A: Screenshot Placeholders

Required Visual Assets

Slide	Asset Needed	Description
4	Architecture Diagram	Detailed technical architecture (sanitized)
5	Executive Dashboard	C-Suite view with strategic KPIs
6	Manager Dashboard	Reference: dtq.digitalworkplace.ai
7	Tech Lead Console	Execution interface with run controls
8	AI Assistant	Chatbot interface showing Q&A
3	Pipeline Animation	Agent workflow visualization
11	Deployment Diagrams	Cloud/GovCloud/On-Prem options

Appendix B: Technical Deep-Dive (Leave-Behind)

For Technical Stakeholders

Knowledge Base Architecture

- Vector database for semantic search
- Structured document ingestion pipeline
- Continuous learning from corrections

Execution Engine

- Playwright MCP integration
- Kubernetes orchestration
- Snapshot manager for page state
- Context-aware prompt generation

Integration Capabilities

- REST API endpoints
- Webhook support
- CI/CD pipeline integration

Note: Detailed technical documentation available under NDA